Lab 3: Central Tendency

Sleep, Test Anxiety, and Memory During Exam Week

2026-08-17

> Student Information
>
> **Name:** ____
> **Date:** ____

Estimated time: 45–60 minutes
Deliverables: completed responses, SPSS output, a saved `.sav` file, and an APA-style results paragraph.

Research scenario

A psychology research team is studying student well-being during exam week. They recruited 24 undergraduate students and recorded each participant's self-reported sleep from the prior night, test-anxiety score, performance on a short memory-recall task, and primary coping strategy.

The team's research question is:

> **During exam week, what are the typical levels of sleep, test anxiety, and memory recall among undergraduate students, and which coping strategy is most common?**

This is a **descriptive** question. Your goal is not to test a hypothesis or compare groups. Instead, you will use the mean, median, and mode to describe the sample appropriately.

Learning objectives

By the end of this lab, you should be able to:

1. Select the mean, median, or mode that best represents a variable.
2. Use SPSS to create frequency tables, descriptive statistics, and histograms.
3. Explain how an extreme score influences the mean and median.
4. Write a concise APA-style descriptive-results paragraph.

Part A — Set up the psychology dataset

Download the psychology lab dataset. In SPSS, select **File > Import Data > CSV Data** (or **File > Open > Data**, depending on your version). Confirm that variable names are read from the first row.

In **Variable View**, enter the value labels and measurement levels below.

Variable	Description / labels	Measure
`id`	Participant identification number	Nominal
`coping_strategy`	1 = Mindfulness; 2 = Planning; 3 = Social support	Nominal
`sleep_hours`	Hours slept the night before the exam	Scale
`test_anxiety`	Test-anxiety score; possible range 0–40, higher = more anxiety	Scale
`memory_recall`	Number of words recalled correctly; possible range 0–20	Scale

Checkpoint A

Save the file as `lab_3_yourlastname.sav`.
Question: Why is `coping_strategy` nominal, even though SPSS displays it using the numbers 1, 2, and 3?
Response:

Part B — The mode: Most Common Coping Strategy

Use **Analyze > Descriptive Statistics > Frequencies**. Move `coping_strategy` to the Variable(s) box. Click **Statistics**, select **Mode**, click continue, then OK. Keep the frequency table in your output.

1. Which coping strategy is the mode? Report its frequency and percentage.

   Response:

2. Why would it be inappropriate to calculate the mean of `coping_strategy`?

   Response:

3. In plain language, write one sentence that answers the coping-strategy portion of the research question.

   Response:

Part C — Typical sleep and memory performance

Use **Analyze > Descriptive Statistics > Explore**. Put `sleep_hours` and `memory_recall` in the **Dependent List**. Click on **Statistics**. Ensure ***Descriptives*** and Percentiles are selected. Under percentiles, select ***Quartiles*** then click ***Continue***. Next, click ***Plots***, then select *Histogram*. Click ***Continue*** the ***OK***. An output should popup.

1. Record the mean and median for each variable.

Variable	Mean	Median
`sleep_hours`		
`memory_recall`		

2. For each variable, describe the histogram as roughly symmetric, positively skewed, or negatively skewed.

   Sleep hours:

   Memory recall:

3. For each variable, decide whether the mean or median is the better single summary of a typical score. Briefly justify your choices.

   Sleep hours:

   Memory recall:

Part D — Test Anxiety and an Extreme Score

Use **Analyze > Descriptive Statistics > Explore** to obtain the mean, median, and histogram for `test_anxiety`. Refer to the steps described in part C above if you need to.

1. Record the mean, median, minimum, and maximum.

Mean	Median	Minimum	Maximum

2. Does the distribution appear skewed? Identify the value that appears to be extreme, and state whether it is low or high relative to the other scores.

   Response:

3. Which measure of central tendency—mean or median—best represents a typical test-anxiety score in this sample? Explain your answer using the effect of the extreme score.

   Response:

Examine the Effect of the Extreme Score

Use **Data > Select Cases**. Choose **If condition is satisfied**, enter `id < 24`, the click continue. Select **Filter out unselected cases**. You will notice that some cases are crossed out. Re-run the Explore analysis for `test_anxiety`.

4. Complete the table. Round to two decimal places.

Cases included	Mean	Median
All 24 cases		
Cases 1–23 only		
Change		

5. Which statistic changed more after the extreme score was excluded? What does this demonstrate about the mean and median?

   Response:

Part E — A simple linear transformation

Before doing this part, we need to Restore **All cases**. To do this, return to **Data > Select Cases > All cases** > **OK**. This will clear the conditional filtering applied earlier.

Now, imagine the research team revises the anxiety scale by adding 5 points to every participant's score. Use **Transform > Compute Variable**. Under ***Target Variable***, write `anxiety_plus_5`. Under Numeric Expression write `test_anxiety + 5`. then click **OK**. A new variable named `anxiety_plus_5` will be created.

Run Explore for `test_anxiety` and `anxiety_plus_5`.

1. Complete the table below:

Variable	Mean	Median
`test_anxiety`		
`anxiety_plus_5`		

2. How did the mean and median anxiety change after adding 5 to each individual score? State the rule for what happens to the mean and median when the same constant is added to every score.

   Response:

Part F — Interpret and report the results in APA style

APA-style results writing should be accurate, concise, and tied to the research question. Use the statistics you produced in SPSS—not estimates from the graph.

APA reporting reminders

- Report the sample size as N when referring to the whole sample.
- Report the mean as M and the standard deviation as *SD*.
- Report the median as *Mdn* when it is the better summary for a skewed distribution.
- Report a categorical mode in words, followed by its frequency (n) and percentage when useful.
- Round consistently, typically to two decimal places unless your instructor directs otherwise.
- Describe the distribution when that explains why you selected the median instead of the mean.

Build your results paragraph

Use this template to write a paragraph that answers the research question:

> The sample ($N =$ ***) reported an average of M =*** hours of sleep ($SD =$ ***; Mdn =***). Participants recalled an average of $M =$ ____ words ($SD =$ ***) on the memory task. The modal coping strategy was [strategy]*** ($n =$, ***%). Test-anxiety scores were [roughly symmetric / positively skewed / negatively skewed]; therefore, the [mean / median] was the most appropriate indicator of a typical score (M or Mdn =***). Overall, these results suggest that [one sentence summarizing the sample descriptively without making a causal claim].

Your APA-style results paragraph

Write a polished 4–6 sentence paragraph below. Use the correct statistical notation and do not state that one variable *caused* another; this study only describes the sample.

Results: ____

Submission checklist

- ☐ A saved SPSS data file with correct value labels and measurement levels.
- ☐ SPSS output containing the frequency table, descriptive statistics, and histograms.
- ☐ Completed responses for Parts A–F. You may print a PDF from the website and hand-write your responses or use a digital annotator tool.

☐ One APA-style results paragraph that directly answers the research question.

Optional challenge

Use **Graphs > Chart Builder** to make a bar chart of `coping_strategy`. Add an informative title and axis labels. In one sentence, explain how the tallest bar communicates the same information as the mode.